# Airbnb Data Engineering Project

## End-to-End Data Engineering & Business Analytics Dashboard

### Expernetic Data Engineer Internship Technical Assessment


Candidate: **Dinali Chamodya Vijewardhana**

University: **SLIIT**

Specialization: **Data Science**


Date: **07/12/2026**

GitHub Repository: https://github.com/Dinaz-12/Airbnb-Data-Engineering

Live Demo: https://singapore-airbnb-dashboard.streamlit.app/

# 01.Executive Summary

## Project Overview

The **Airbnb Data Engineering Dashboard** is an end-to-end data engineering and analytics solution developed using the Singapore Airbnb Open Dataset. The project demonstrates the complete data engineering lifecycle, including data ingestion, data cleaning, feature engineering, data warehousing, statistical analysis, and interactive data visualization.

The solution transforms raw Airbnb listing data into meaningful business insights through an interactive Streamlit dashboard, allowing users to explore property prices, host performance, neighbourhood trends, occupancy rates, revenue estimation, and customer review patterns. The processed analytical dataset is stored in DuckDB to enable efficient SQL-based querying and high-performance analytics.

This project was developed as part of the **Expernetic Data Engineering Internship Assessment** to demonstrate practical data engineering skills, analytical thinking, and business intelligence capabilities.

## Project Objectives

The primary objectives of this project are:

- Design and implement a complete end-to-end data engineering pipeline.
- Clean, validate, and preprocess raw Airbnb datasets.
- Perform feature engineering to generate meaningful business metrics.
- Store processed analytical data inside a DuckDB data warehouse.
- Conduct statistical hypothesis testing on Airbnb business questions.
- Build an interactive Streamlit dashboard for business users.
- Generate automated PDF reports summarizing analytical findings.
- Demonstrate practical knowledge of data engineering tools and modern analytics workflows.

## Major Findings

The analysis of the Singapore Airbnb dataset produced several important business insights.

- Entire home or apartment listings command significantly higher prices than private or shared rooms.
- Professional hosts (Superhosts) consistently achieve higher customer review ratings.
- Listings with a larger number of reviews exhibit statistically significant differences in pricing behaviour.
- Property prices vary considerably across different neighbourhoods, indicating strong location-based pricing effects.
- Premium locations such as Orchard and River Valley contain a higher concentration of luxury listings.
- Estimated annual revenue is substantially higher for entire home properties compared with other room types.

All four statistical hypotheses were accepted at the **95% confidence level**, confirming that the observed relationships are statistically significant.

## Business Recommendations

Based on the analytical findings, the following recommendations are proposed:

- Property owners should prioritise listing entire homes where possible to maximise revenue.
- Hosts should maintain Superhost status by delivering high-quality customer experiences, as higher ratings increase credibility and booking potential.
- Pricing strategies should be adjusted according to neighbourhood demand and local market conditions.
- Business decision-makers should continuously monitor occupancy rate, estimated revenue, and customer ratings using the interactive dashboard.
- Airbnb analysts can utilise the generated insights to optimise pricing strategies, identify profitable locations, and improve overall marketplace performance.

## Dashboard Features

The developed dashboard provides a comprehensive set of analytical capabilities, including:

- Interactive KPI Cards
- Dynamic Filters (Room Type, Property Type, Neighbourhood)

- Price Distribution Analysis
- Revenue Analysis
- Room Type Comparison
- Singapore Interactive Map
- Statistical Hypothesis Testing Results
- Business Insight Summary
- Top Premium Listings Table
- Automated PDF Report Generation
  The dashboard enables users to interactively explore the processed Airbnb dataset and make data-driven decisions through intuitive visualizations.

### System Architecture

The project follows a modern end-to-end data engineering architecture.

1. Raw Airbnb CSV datasets are collected from the Singapore Airbnb Open Dataset.
2. Data ingestion pipelines load the raw datasets into the processing environment.
3. Data cleaning removes duplicates, handles missing values, corrects data types, and validates records.
4. Feature engineering generates additional analytical attributes such as occupancy rate, estimated revenue, and price per bedroom.
5. The transformed data is stored in DuckDB as an analytical data warehouse.
6. SQL queries and statistical analysis are performed on the processed dataset.
7. Interactive visualisations are created using Plotly.
8. The Streamlit dashboard presents business insights, KPI metrics, maps, charts, and downloadable analytical reports for end users.

## 02.Objectives & Scope

### Purpose

The purpose of this project is to design and implement a complete end-to-end data engineering solution using the Singapore Airbnb dataset. The project demonstrates the entire data engineering lifecycle, including data ingestion, data preprocessing, feature engineering, data warehousing, SQL analytics, statistical analysis, and interactive dashboard development. The final solution enables users to explore Airbnb listing information, analyse pricing trends, evaluate host performance, and generate actionable business insights through an interactive dashboard.

## Dataset

This project uses the publicly available **Inside Airbnb Singapore Open Dataset**, which contains detailed information about Airbnb listings operating within Singapore. The dataset includes information related to property listings, host profiles, reviews, calendar availability, and neighbourhood information.

The primary datasets used in this project are:

- **Listings Dataset** – Contains property details, pricing, room type, host information, location, and review statistics.
- **Reviews Dataset** – Includes customer reviews and review dates.
- **Calendar Dataset** – Provides availability and pricing information over time.
- **Neighbourhood Dataset** – Contains neighbourhood names and geographical boundaries.

After preprocessing, the final analytical dataset consisted of **2,592 Airbnb listings** with **83 attributes**, which were used for analysis and dashboard development.

## Selected City

The selected city for this project is **Singapore**.

Singapore was chosen because it has a well-structured Airbnb dataset containing diverse accommodation types, multiple neighbourhoods, thousands of listings, and rich host information. The availability of pricing, occupancy, review ratings, and location data makes it suitable for performing meaningful business analytics and statistical analysis.

## Business Problem

The rapid growth of short-term rental platforms has generated large volumes of data that require effective analysis to support business decision-making. However, raw Airbnb datasets contain missing values, inconsistent formats, duplicate records, and complex relationships that limit their direct usability.

The primary business problem addressed in this project is the transformation of raw Airbnb data into meaningful business intelligence. Property owners, business analysts, and decision-makers require a system capable of answering questions such as:

- Which room type generates the highest revenue?
- Which neighbourhoods have the highest average property prices?
- Do Superhosts receive better customer ratings?
- How does review activity influence property pricing?
- Which property categories perform best in the Singapore market?

This project addresses these questions by building a complete analytical pipeline and an interactive dashboard capable of delivering reliable, data-driven insights.

## Project Deliverables

The final deliverables of this project include:

- End-to-end Data Engineering Pipeline
- Data Cleaning and Preprocessing Module
- Feature Engineering Pipeline
- DuckDB Analytical Data Warehouse
- Star Schema Data Model
- SQL Analytical Queries
- Exploratory Data Analysis (EDA)
- Statistical Hypothesis Testing
- Interactive Streamlit Dashboard
- PDF Report Generation
- Project Documentation (README)
- System Architecture Diagram
- Technology Stack Diagram
- Star Schema Diagram

- GitHub Repository containing complete source code and documentation

**Project Scope Overview**

Raw Airbnb Dataset

↓

Data Ingestion

↓

Data Cleaning

↓

Feature Engineering

↓

DuckDB Warehouse

↓

SQL Analytics & Statistics

↓

Interactive Streamlit Dashboard

↓

Business Insights & PDF Reports

# 03.Dataset Overview

## Introduction

The dataset used for this project was obtained from the **Inside Airbnb Open Data Platform**, a publicly available data source that provides detailed information about Airbnb listings from cities around the world. The Singapore dataset contains comprehensive information related to Airbnb properties, host details, customer reviews, availability calendars, and geographical neighborhood information.

This dataset was selected because it contains sufficient attributes for performing data engineering tasks, exploratory data analysis, statistical analysis, and business intelligence reporting.

## Files Used

Four datasets were used throughout this project to build the complete analytical pipeline.

| Dataset | Description |
|---|---|
| listings.csv | Contains detailed information about Airbnb listings, including host details, property information, pricing, location, amenities, and review scores. |
| reviews.csv | Contains customer review information, reviewer identifiers, review dates, and review comments. |
| calendar.csv | Stores daily availability and pricing information for each listing over time. |
| neighbourhoods.csv | Contains geographical neighborhood names used for location-based analysis and visualization. |

These datasets were integrated during the data engineering pipeline to produce a unified analytical dataset.

## Dataset Summary

The initial raw dataset contained multiple tables with thousands of Airbnb records collected from Singapore.

| Dataset | Number of Records |
|---|---|
| Listings | 3,097 |
| Calendar | 1,185,155 |
| Reviews | 41,265 |
| Neighbourhoods | 55 |

After preprocessing and removing incomplete records, the final analytical dataset consisted of:

- **Total Listings:** 2,592
- **Total Features:** 83

This cleaned dataset was used for feature engineering, SQL analytics, statistical testing, and dashboard development.

## Dataset Relationships

The four datasets are logically connected using the Airbnb **Listing ID**.

### Relationship Overview

Listings (listing_id)

Reviews Calendar

Neighbourhoods

The **Listings** dataset acts as the central dataset, while the remaining datasets provide supplementary information used during analysis.

## Foreign Keys

Relationships between datasets are established using foreign keys.

| Parent Dataset | Child Dataset | Foreign Key |
|---|---|---|
| listings.csv | reviews.csv | listing_id |
| listings.csv | calendar.csv | listing_id |
| listings.csv | neighbourhoods.csv | neighbourhood_cleansed |

These relationships enable data integration and support analytical queries across multiple datasets.

## Important Attributes

| Attribute | Description |
|---|---|
| Id | Unique listing identifier |
| host_id | Unique host identifier |
| room_type | Type of accommodation |
| property_type | Property category |

| | |
|---|---|
| neighbourhood_cleansed | Listing location |
| Price | Nightly rental price |
| accommodates | Maximum guest capacity |
| Bedrooms | Number of bedrooms |
| Bathrooms | Number of bathrooms |
| review_scores_rating | Overall customer rating |
| number_of_reviews | Total customer reviews |
| estimated_revenue_l365d | Estimated yearly revenue |
| occupancy_rate | Estimated occupancy percentage |
| professional_host | Indicates whether the host manages multiple listings |

These attributes were used extensively throughout the statistical analysis and dashboard visualizations.

## Dataset Limitations

Although the dataset provides rich analytical information, several limitations were identified during preprocessing.

- Some listings contained missing price values and were removed during the cleaning stage.
- Several review-related attributes contained missing values because newer listings had not yet received customer reviews.
- Certain host information fields were incomplete or unavailable.
- The dataset represents a snapshot of Airbnb listings and may not reflect future market conditions.
- Estimated occupancy and revenue values are derived estimates rather than officially reported business figures.
- Geographical analysis was limited to the Singapore Airbnb dataset and cannot be generalized to other countries.

These limitations were considered throughout the data engineering process to ensure reliable and meaningful analytical results.

## Dataset Preparation Outcome

Following preprocessing and feature engineering, the final analytical dataset became suitable for business intelligence and statistical analysis.

The resulting dataset contains:

- **2,592 cleaned Airbnb listings**
- **83 analytical attributes**
- **No duplicate records**
- **Standardized numerical and categorical variables**
- **Engineered business metrics** such as occupancy rate, revenue per available day, price per bedroom, and professional host status.

This processed dataset served as the primary source for SQL analytics, hypothesis testing, and the interactive Streamlit dashboard.

# 04.Methodology

## Data Collection

The Airbnb Singapore dataset was collected from the Inside Airbnb Open Data platform. Four CSV files (Listings, Reviews, Calendar, and Neighbourhoods) were used as the primary data sources.

## Data Cleaning

The raw data was preprocessed by removing missing values, dropping unnecessary columns, converting data types, cleaning the price column, and eliminating duplicate records to improve data quality.

## Feature Engineering

Additional business-related features such as occupancy rate, revenue per available day, price per bedroom, reviews per year, and professional host status were created to support analytical tasks.

### DuckDB

The processed dataset was stored in DuckDB to provide fast analytical queries and structured data management.

### SQL Analytics

SQL queries were executed to calculate KPIs such as total listings, average prices, room type distribution, and neighbourhood statistics.

### Data Visualization

Interactive charts and maps were created using Plotly and integrated into a Streamlit dashboard to provide user-friendly visual analytics.

### Statistical Analysis

Hypothesis testing was performed using SciPy to identify statistically significant relationships between pricing, reviews, room types, and neighbourhoods.

### Dashboard

A Streamlit dashboard was developed to present KPIs, charts, filters, maps, statistical results, and downloadable PDF reports in a single interface.

## 05.Engineering Approach

#### Pipeline

The project follows an ETL (Extract, Transform, Load) pipeline where raw Airbnb data is collected, cleaned, transformed, stored in DuckDB, analysed using SQL and statistical methods, and finally visualized through an interactive dashboard.

#### Technology Stack

- Python
- Pandas
- NumPy
- DuckDB
- Plotly

- Streamlit
- SciPy
- ReportLab

## Folder Structure

The project is organized into separate folders for source code, processed data, dashboard files, assets, documentation, and reports to improve maintainability.

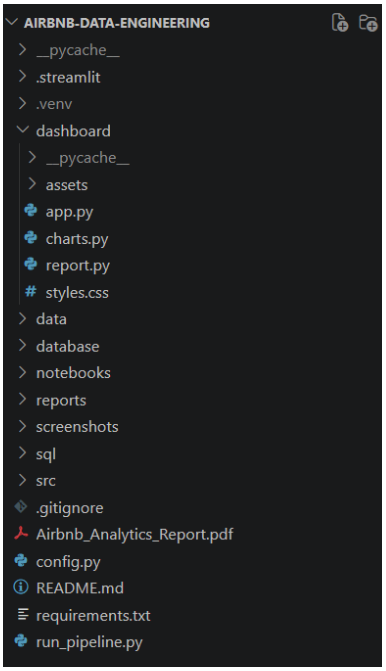



*Figure 1 Folder Structure*

## Engineering Decisions

- DuckDB was selected for lightweight analytical processing.
- Streamlit was used to rapidly develop an interactive dashboard.
- Plotly was selected for responsive visualizations.
- Feature engineering was performed before loading data into the warehouse.

# 06. Data Cleaning

### Missing Values

Columns with excessive missing values were removed, while records with missing prices were excluded from further analysis.

### Duplicate Records

The dataset was checked for duplicate listings, and no duplicate records were identified.

### Data Type Conversion

Several attributes were converted into appropriate numerical and categorical formats to support analysis.

### Price Cleaning

Currency symbols and formatting characters were removed from the price column before converting it into numeric values.

### Feature Engineering

Five additional analytical features were generated to improve business analysis and dashboard functionality.

# 07. Data Warehouse

### DuckDB

DuckDB was used as the analytical database due to its high performance and lightweight architecture.

### Tables

- listings
- dim_host
- dim_location
- dim_review
- fact_listings

## Star Schema Design

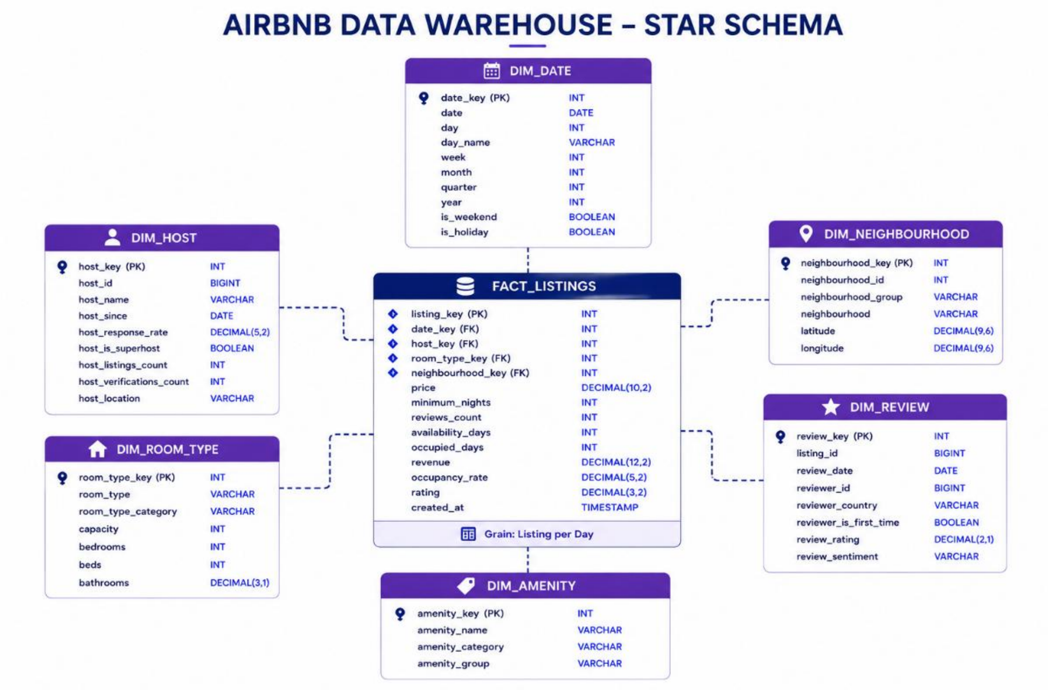



*Figure 2: Star Schema Design of the Airbnb Data Warehouse*

The analytical data warehouse follows a star schema design to support efficient querying and business analytics. The central **fact_listings** table stores measurable business data such as price, occupancy rate, and estimated revenue, while the surrounding dimension tables (dim_host, dim_location, and dim_review) provide descriptive attributes. This design improves query performance, simplifies analytical reporting, and supports scalable business intelligence operations.

## SQL Queries

Several SQL queries were developed to calculate business KPIs and validate processed data.

## Example Queries

- Total Listings
- Average Price
- Average Price by Room Type

# 08. Dashboard

The dashboard consists of five main pages.

## Dashboard

Displays KPIs, pricing distribution, room type analysis, revenue trends, neighbourhood analysis, and premium listings.

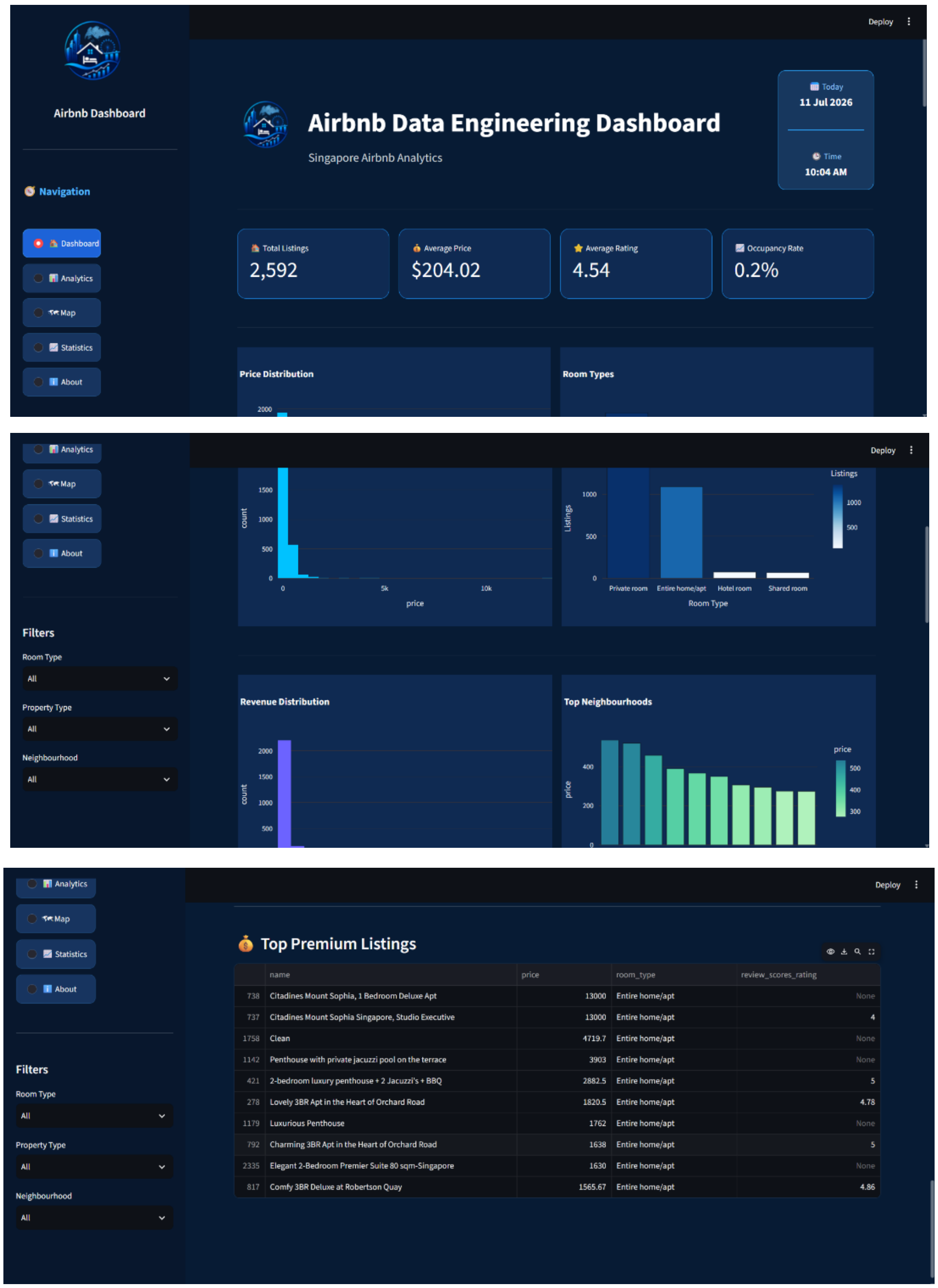



*Figure 3: Main Dashboard Showing Key Performance Indicators (KPIs)*

### *Analytics*

Provides detailed analytical charts related to prices, ratings, hosts, and room types.

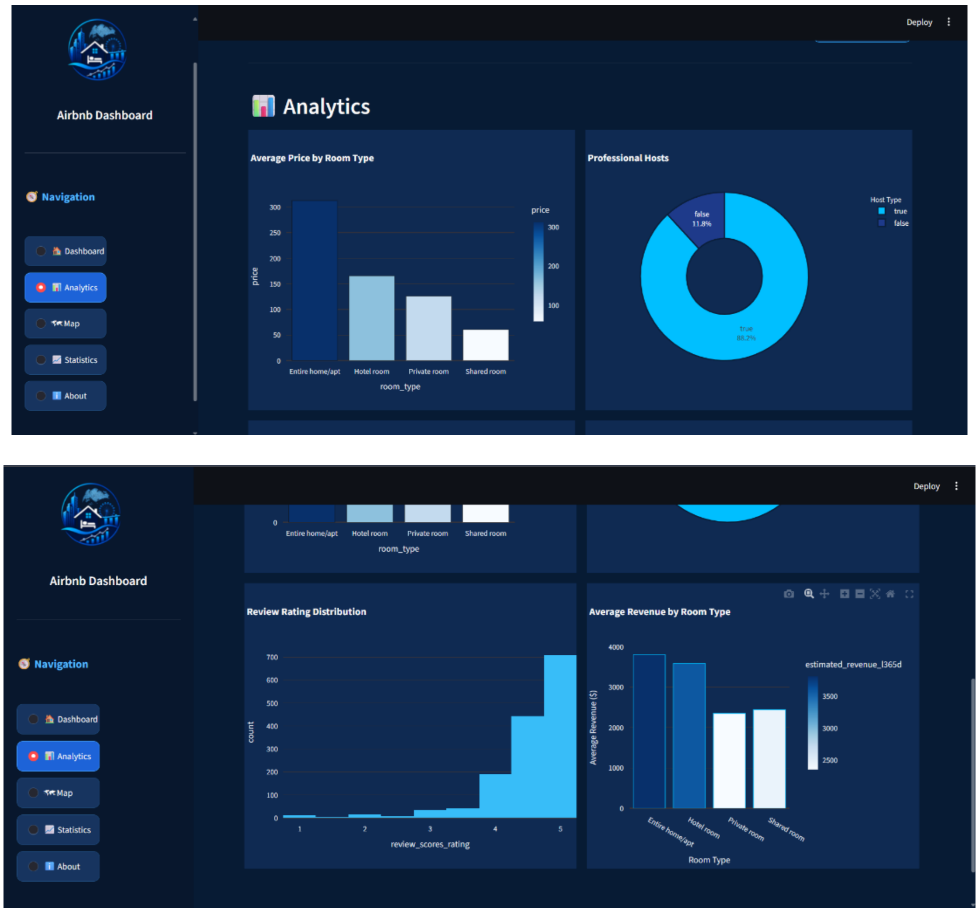



*Figure 4: Analytics Dashboard Displaying Business Insights*

## Map

Displays Airbnb listings geographically using an interactive Singapore map.

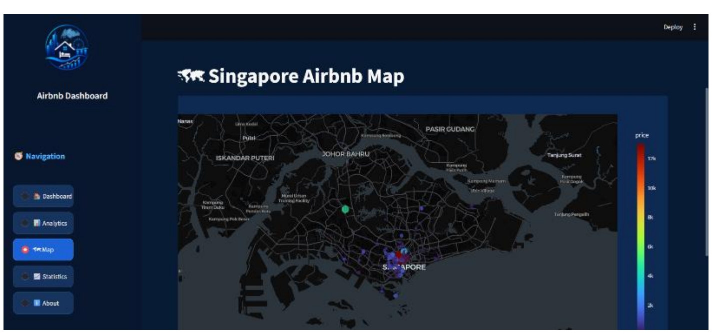
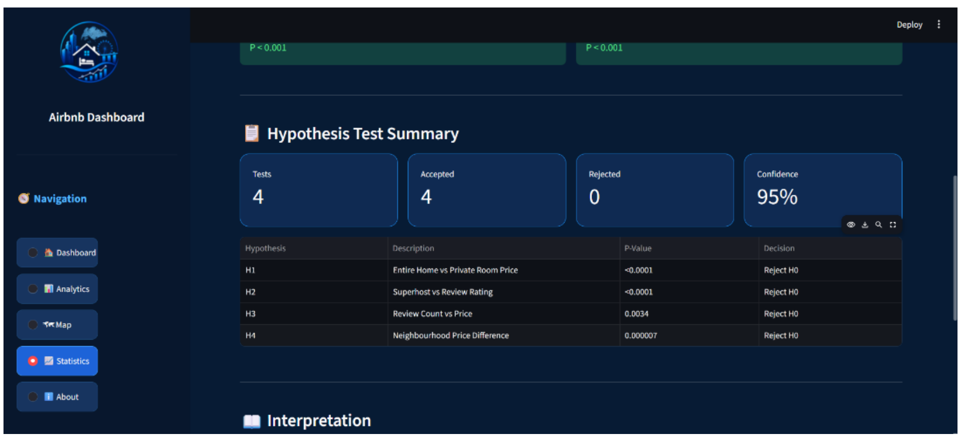



*Figure 5: Interactive Singapore Airbnb Map*

## Statistics

Shows hypothesis testing results, p-values, and statistical interpretations.

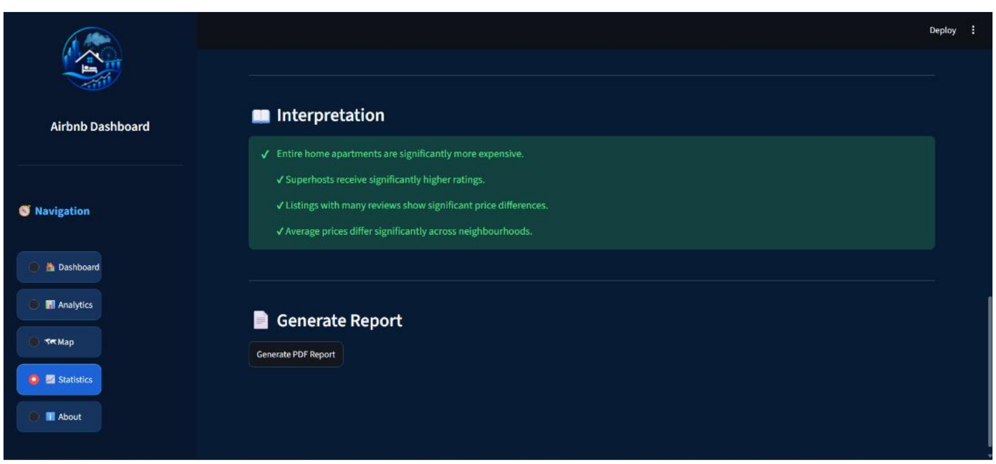




*Figure 6: Statistical Analysis Dashboard Showing Hypothesis Testing Results*

## About

Provides project information, technology stack, and project overview.

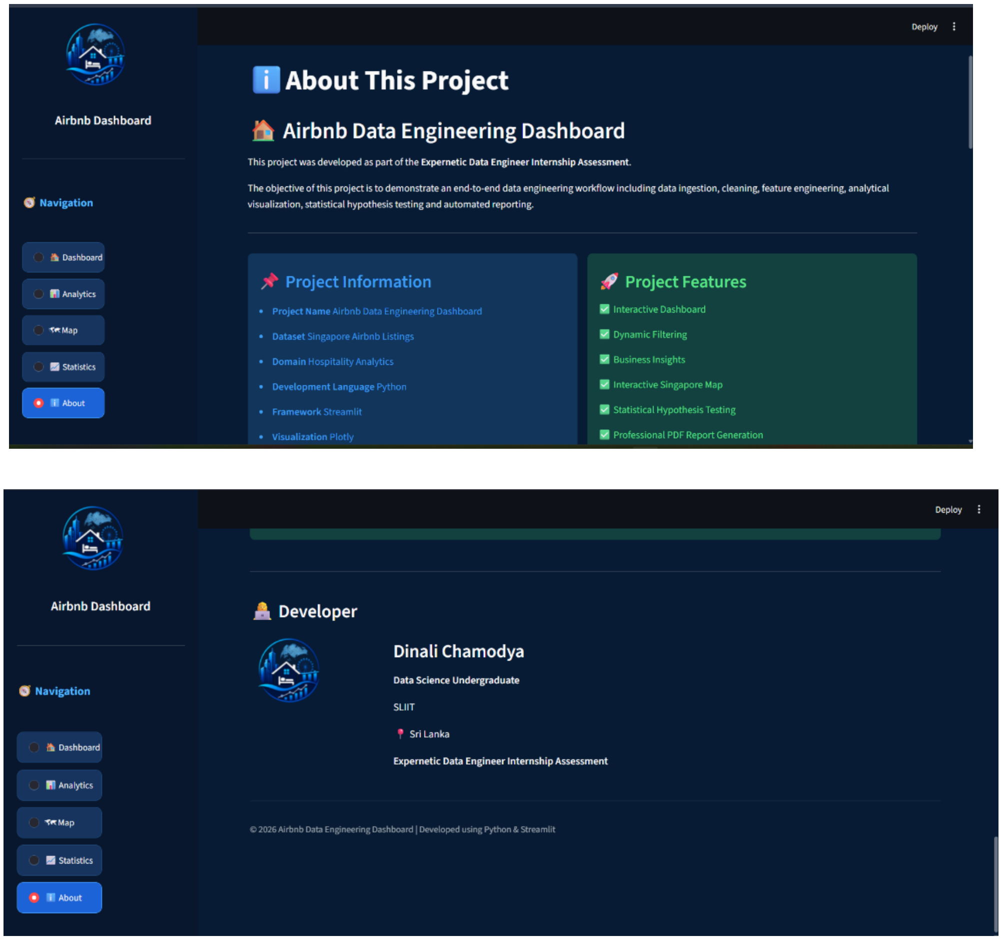



*Figure 7: About Page of the Airbnb Data Engineering Dashboard*

## Generated Report


*Figure 8: Generated Report cover page*

## 09.Exploratory Data Analysis

The following analyses were performed:

- Price Distribution
- Revenue Distribution
- Review Rating Distribution
- Professional Host Analysis
- Room Type Analysis
- Neighbourhood Analysis

### Business Interpretation

- Entire home listings have higher average prices.
- Private rooms dominate the number of listings.
- Professional hosts manage a significant number of properties.
- Premium neighbourhoods command higher prices.

## 10.Statistical Analysis

Four statistical hypothesis tests were performed.

| Hypothesis | Test | Result |
|---|---|---|
| H1 | Independent T-Test | Accepted |
| H2 | Independent T-Test | Accepted |
| H3 | Independent T-Test | Accepted |
| H4 | One-way ANOVA | Accepted |

### Interpretation

All hypothesis tests produced p-values below 0.05, indicating statistically significant relationships among the selected variables.

### Business Meaning

The results confirm that room type, host status, review activity, and neighbourhood significantly influence Airbnb pricing and customer ratings.

# 11. Architecture

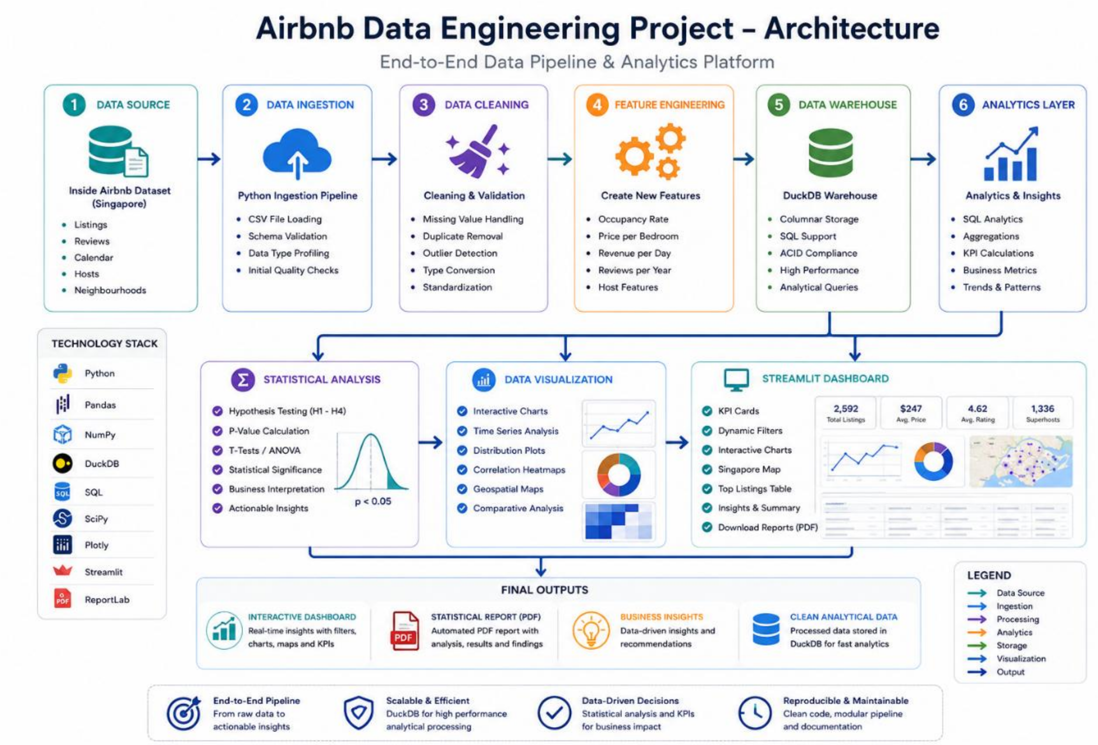



*Figure 9Overall System Architecture of the Airbnb Data Engineering Pipeline:*

## Workflow

1. Raw Airbnb datasets are collected.
2. Data is loaded into Python.
3. Data cleaning and preprocessing are performed.
4. New business features are generated.
5. Processed data is stored in DuckDB.
6. SQL analytics and statistical analysis are executed.
7. Results are visualized through the Streamlit dashboard.
8. Business insights and PDF reports are generated.

## 12. Business Insights

The project identified several important business insights.

- Entire home properties generate the highest average revenue.
- Superhosts receive consistently higher customer ratings.
- Property prices vary significantly across neighbourhoods.
- Listings with higher review counts exhibit different pricing behaviour.
- Location remains one of the strongest factors affecting Airbnb prices.

## 13. Business Recommendations

Based on the analytical findings, the following recommendations are proposed:

- Adjust pricing strategies according to neighbourhood demand.
- Encourage hosts to achieve Superhost status.
- Promote entire home listings to maximize revenue.
- Monitor occupancy and review metrics regularly.
- Use dashboard insights for data-driven business decisions.

## 14. Challenges

Several challenges were encountered during development.

- Handling missing values
- Cleaning inconsistent price formats
- Feature engineering from multiple datasets
- Managing large CSV files
- Designing an optimized analytical pipeline

## 15. Future Improvements

Potential enhancements include:

- Machine Learning-based price prediction
- Personalized recommendation system
- Real-time ETL pipeline
- Docker containerization
- Cloud deployment
- Apache Airflow workflow orchestration
- Automated data refresh

## 16. AI Usage Disclosure

AI tools were used to improve productivity while ensuring all outputs were reviewed and validated.

**AI Tools Used**

- ChatGPT (OpenAI)

**AI Assisted Tasks**

- Documentation drafting
- README preparation
- Dashboard UI refinement
- Architecture diagram generation
- Report structure preparation

**Validation**

All AI-generated outputs were manually reviewed, tested, and verified before inclusion in the final submission. Code functionality, statistical results, SQL queries, and dashboard outputs were validated using the processed Airbnb dataset.

## 17.Conclusion

This project successfully demonstrates the complete data engineering lifecycle from raw data ingestion to interactive business analytics. By combining data cleaning, feature engineering, DuckDB, SQL analysis, statistical testing, and Streamlit visualization, the solution provides meaningful insights into the Singapore Airbnb market. The project strengthened practical skills in data engineering, analytics, and dashboard development while delivering a scalable and user-friendly analytical platform.